

Entregável 3.2. Demonstrador prático: mapas nacionais vetoriais a delimitar manchas de perda recente de floresta e mato superiores a 0.5 ha com uma frequência bimestral relativos a um período contínuo de dois anos entre 2023 e 2025.

1. Mapas nacionais vetoriais disponíveis para descarregamento

Mapas vetoriais bimestrais para os anos 2023 e 2024 foram elaborados como resultado de processamento de séries temporais Sentinel-2 através do algoritmo CCD para Portugal Continental e para a máscara de potenciais perdas de vegetação fornecida pela DGT (aproximadamente 500 M pixels Sentinel-2 de resolução 10 m).

O processamento foi realizado em ambiente HPC usando a plataforma INCD/CNCA (Cirrus).

Os mapas (em formato shapefile) estão disponíveis em https://github.com/S2change/vegetation_loss/tree/main/data_info/vegetation_loss_products

No total foram criados 4 produtos, um completo e disponibilizado (MBPV_v1: *mapa bimestral de perdas de vegetação v1*), e 3 outros produtos experimentais: CCD_RPV_v2, CCD_MIPV_v2, e CCD_MBPV_v2.

2. Descrição dos produtos de perdas de vegetação (em inglês)

1. CCD_RPV_v2: Raster product. Similarly to MBPV_v1, the Sentinel-2 series is processed up to a recent date (e.g. present), but the vector vegetation loss map is computed for some time interval I. For instance, S2 is processed by PyCCD up to 2025-11-17, but only breaks within ['2023-01-01','2023-02-28'] are extracted to create the vector product. Processing steps are similar to MBPV_v1 with a few improvements.
2. CCD_MIPV_v2: (mapa de perdas de vegetação v2). Similarly to MBPV_v1, the Sentinel-2 series is processed up to a recent date (e.g. present), but the vector vegetation loss map is computed for some time interval I. For instance, S2 is processed by PyCCD up to 2025-11-17, but only breaks within ['2023-01-01','2023-02-28'] are extracted to create the vector product. Processing steps are similar to MBPV_v1 with a few improvements.

3. CCD_MBPV_v2: (near-real time mapa bimestral de perdas de vegetação v2). Instead of being based on a Sentinel-2 time series up to 2024-12-31 (as MBVP_v1), this product is a near-real time bymonthly product, where for each date range, e.g. ['2023-01-01','2023-02-28'], the Sentinel-2 time series is only processed up to the last date of the data range. Otherwise, the processing steps are similar to MBVP_v1 with a few improvements.
4. MBPV_v1: (mapa bimestral de perdas de vegetação v1). This is an experimental product for 2023-2024 that is solely based on the PyCCD estimated breaks. Furthermore, the PyCCD processing uses a parameter (minimum number of observations required to identify break) which is probably too high and makes it harder to determine breaks, in particular at the end of the temporal series. Neighbor pixels with similar dates (less than 10 days) are clustered and clusters with area smaller than 0.5 ha are discarded.

The steps to produce MBPV_v1 are the following:

- Process Sentinel-2 time series up to 2024-12-31 with PyCCD.
Input. hdf5 file with 4 bands B3, B4, B8, B12 (NA=65355), CRS=32629 + hdf5 files for x,y, DGT vegetation loss mask (see /data_info/readme.md)
Outputs: Parquets PyCCD outputs (see /data_info/readme.md)
- Convert PyCCD output to raster files with the most recent detection date (script at /scripts/visualisations/ccd_to_raster.py)
Input: Parquets PyCCD outputs; date_range, e.g. ['2023-01-01','2023-02-28'], so only breaks within the date range are considered. For this product, date_range is bimonthly, and one segment from CCD is always assigned to a single bimonthly period.
Output: Multi-band GeoTIFF raster file (.tif), with bands last_tEnd, last_tBreak, is_break, ndvi_last_segment: see /scripts/visualisations/ccd_to_raster.py for details.
- Convert each bimonthly raster file to vector format with script /scripts/visualisations/graph_raster_to_polygons.py:
Input: For each period (2 months) and for each tile: 1 multi-band GeoTIFF raster file (.tif); date is the mean between last_tEnd and last_tBreak, date_range_days = 10 (number of days to group adjacent pixels within each spatial cluster); min_area_ha = 0.5 (minimum polygon area in hectares).
Output: For each period (2 months) and for each tile: 1 polygon geopackage file; attributes: mean, min, max dates for each polygon; polygon area, with polygons at least 0.5 ha.